DIGITAL FORENSICS PLAYBOOK

Handling and Preserving Digital Evidence

Focus	Evidence integrity, documentation, and responsible handling
Scenario	Hypothetical ransomware incident
Project type	Educational digital forensics documentation project

Purpose

This playbook organizes foundational practices for identifying, preserving, collecting, examining, analyzing, and presenting digital evidence. It emphasizes evidence integrity, documentation of handling activities, and protection of sensitive information throughout an investigation.

Scope note
This is an educational reference based on a hypothetical scenario. It is not a report of a live incident and does not claim that a real ransomware investigation was conducted. Procedures must be adapted to applicable law, organizational policy, and qualified incident-response guidance.

1. Digital Forensics Lifecycle

A structured process helps preserve the reliability of evidence and communicate findings clearly. These stages provide a practical framework; real investigations may require revisiting earlier stages as new information becomes available.

1. Identification

Identify potential sources of relevant evidence, such as computers, mobile devices, external drives, cloud services, and Internet of Things (IoT) devices. Record what was found and its context.

2. Preservation

Protect evidence from unnecessary alteration or loss. Consider the device's condition, possible volatile evidence, and appropriate steps for preserving the source and establishing chain of custody.

3. Collection

Acquire relevant data using methods appropriate to the source. Potential sources include forensic images, system logs, memory, network records, authentication records, backups, and cloud data.

4. Examination

Inspect collected material—such as files, logs, metadata, memory artifacts, and other records—to identify information relevant to the investigation.

5. Analysis

Interpret examined artifacts to develop timelines, identify relationships, assess likely causes, and document supported conclusions and limitations.

6. Presentation

Prepare a clear report describing methods, evidence, findings, limitations, and recommendations for technical and non-technical audiences.

2. Hypothetical Scenario: Ransomware

A ransomware incident may affect endpoints, accounts, network resources, logs, backups, and cloud services. Evidence handling should support incident response and later review while reducing the risk of losing or altering relevant artifacts.

Identification and initial scoping

- Identify potentially affected systems, users, services, and available evidence sources.
- Record observable symptoms, known timestamps, system identifiers, and actions already taken.
- Consider likely initial access and whether additional systems or accounts may be affected.

Preservation and collection

- Follow the organization's incident-response process when deciding how to contain an affected system and preserve evidence.
- Consider whether volatile information—such as running processes, active network connections, logged-in users, open files, and memory—may be relevant before a system is shut down or changed.
- Protect relevant logs from loss or alteration and document the source, collection time, method, and person responsible.
- Where appropriate and authorized, acquire forensic images and collect relevant system, network, authentication, backup, and cloud records.

Examination and analysis

- Review relevant files, logs, processes, accounts, and network artifacts.
- Build a timeline from available evidence and document the basis for each conclusion.
- Assess observed activity and potential impact; note gaps or uncertainties rather than treating assumptions as facts.

Presentation

- Summarize scope, methods, evidence sources, key findings, limitations, and recommended follow-up.
- Tie conclusions to documented artifacts and distinguish confirmed observations from hypotheses.

3. Handling and Preserving Digital Evidence

Documentation begins before acquisition and continues whenever evidence is collected, transferred, stored, accessed, or examined.

Document the source device

- Record device type, manufacturer, model, serial number when available, storage capacity, physical condition, and a unique evidence identifier.
- Photograph or otherwise document the device and its condition when appropriate.
- Record date and time, time zone where relevant, location, collector, and circumstances of collection.

Avoid unnecessary changes to original media

Connecting storage media to a general-purpose computer can alter data or metadata. Where appropriate, use suitable write-blocking methods and established forensic procedures to reduce the risk of modifying original evidence. Document the methods and tools used.

Forensic imaging

A forensic image is a copy of storage contents created for examination. The general objective is to preserve the original source while examining an acquired copy. Depending on the source and acquisition method, an image may include active files, deleted-file remnants, file-system metadata, unallocated space, and other structures.

Preferred workflow
Original source → **Forensic acquisition** → **Forensic image** → **Examination**

Protect the acquired evidence

- Restrict access to authorized personnel.
- Use encryption and controlled storage for sensitive evidence when appropriate.
- Maintain audit records of access and handling activities.
- Maintain protected backups when permitted by policy and evidence requirements.
- Document acquisition details, including examiner, date/time, evidence identifier, source device, method, tools, write-blocking method if used, image format, and image size.

4. Integrity and Hashing

Cryptographic hash values can help assess whether a digital file or image has changed between comparison points. Record the algorithm and hash value, the item to which it applies, and when and how the value was calculated. Hash agreement supports an integrity check but does not, by itself, establish the origin, completeness, or meaning of evidence.

5. Chain of Custody

A chain-of-custody record documents who collected, possessed, transferred, stored, or examined an evidence item and when those events occurred. Clear records help others evaluate how evidence was handled over time.

A useful record should include:

- Evidence identification and a unique item number.

- Collection details, including who collected the item, when, where, and by what method.
- Each transfer or change of custody, with date/time, releasing and receiving parties, and purpose.
- Storage location and relevant access or examination events.
- Signatures or other approved acknowledgments under organizational procedures.

6. Legal, Ethical, and Reporting Considerations

Digital evidence should be collected and examined only under appropriate legal authority and within the authorized scope. Investigators should avoid unnecessary access to unrelated personal information and follow applicable law, organizational policy, and professional procedures.

Core principles

- **Integrity:** Use appropriate methods to protect evidence from unnecessary alteration and document any issues.
- **Accountability:** Maintain accurate records of collection, transfer, storage, access, and examination.
- **Confidentiality:** Protect sensitive information and limit access to authorized people.
- **Objectivity:** Report what the evidence supports, including uncertainty, limitations, and alternative explanations where relevant.
- **Transparency:** Describe methods and tools used so the work can be understood and evaluated.

Reporting checklist

Item	Questions to address
Scope	What was examined, and what authority or scope applied?
Methods	What collection and examination methods and tools were used?
Evidence	Which artifacts support each important observation?
Integrity	What preservation, hashing, and custody records are available?
Findings	What can be concluded, and what remains uncertain?
Next steps	What follow-up, containment, recovery, or review is recommended?

7. Key Takeaways

- Use a documented lifecycle to organize evidence-handling work.
- Preserve source data and record the steps used to collect and examine it.
- Use hashes as one component of integrity verification, alongside sound procedures and clear records.
- Maintain a traceable chain of custody and secure evidence against unauthorized access.
- Keep reporting factual, impartial, and clear about limitations.

Portfolio note: This document is an educational summary of digital forensics concepts and a hypothetical ransomware scenario. It is not legal advice or an operational runbook for an active incident. Follow qualified incident-response guidance and applicable organizational procedures.